TECHNICAL & COMMERCIAL PROPOSAL

PERMITTO

Permit To Work (PTW) Digital Management System

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[Your Company Name]

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Project Introduction

Project Background

PowerChina HDEC currently manages its Permit To Work (PTW) process through paper-based forms and manual approval workflows. While this approach has served operational needs, the increasing scale of projects, the need for real-time visibility across multiple zones, and the requirement for HSSE (Health, Safety, Security & Environment) compliance create significant challenges in managing permits efficiently and safely.

Paper-based permits are susceptible to delays in the approval chain, lost or incomplete forms, difficulty in tracking simultaneous operations (SIMOPs), and the inability to provide real-time status visibility to all stakeholders. The daily re-approval cycle — a critical safety requirement — is particularly burdensome to manage manually across multiple active permits and zones.

This proposal presents Permitto, a purpose-built digital Permit To Work management system designed to address these challenges by digitalizing PowerChina HDEC's existing PTW process while maintaining full compliance with current operational procedures.

Project Objectives

- **Digitalize the PTW Process —** Replace paper-based permit forms with a structured digital workflow that mirrors the existing approval chain (WPR → WPI → PTW-PC → HSSE)
- **Ensure HSSE Compliance —** Embed the HSSE Officer as a formal approval level in every permit, with independent risk assessment verification and the ability to request additional safety measures
- **Manage Simultaneous Operations —** Automate SIMOPs detection, enforce countersign requirements between concurrent operations in the same zone, and provide a withholding mechanism until interface management is resolved
- **Enable Real-Time Visibility —** Provide a centralized PTW Register with real-time status tracking, searchable and filterable by project, zone, permit type, and status
- **Automate Daily Re-Approval —** Digitalize the daily re-confirmation cycle with deadline enforcement, automatic suspension of permits with missed approvals, and bulk approval capability
- **Support Field Operations —** Deliver a Progressive Web App (PWA) with secure offline mode, allowing field workers to view permits, fill forms, and queue approvals even without connectivity
- **Reduce Administrative Overhead —** Automate notifications, audit logging, PDF generation, and status transitions to minimize manual administrative work
- **Deploy Self-Contained —** Deliver a complete system that runs entirely on the client's VPS with no external cloud service dependencies or recurring infrastructure costs

Solution Overview

Permitto is a web-based application built with modern, proven technologies. The system architecture is designed for reliability, performance, and ease of deployment on a single VPS server.

Component	Technology	Purpose
Frontend	Next.js (React)	Responsive web application with PWA support
Backend	Express.js (Node.js)	REST API with real-time processing
Database	PostgreSQL	Primary data store — permits, users, signatures, audit logs
Cache & Queue	Redis	Session management, background job queues
PDF Engine	Puppeteer (server-side)	On-demand permit PDF generation
Deployment	Docker	Containerized deployment for consistency and reliability
File Storage	VPS Filesystem	Supporting documents stored securely on server

Key Design Principle: Zero External Dependencies

The entire system — application, database, cache, file storage, and PDF generation — runs self-contained on the client's VPS. There are no external cloud storage fees (R2, S3), no third-party PDF storage, and no recurring infrastructure costs beyond the server itself and the email service provided by the client.

Document Structure

This proposal is organized into the following sections:

Section	Title	Description
Section 1	Scope of Work	Defines all in-scope deliverables, the permit workflow, out-of-scope items, and project assumptions
Section 2	System Features & Modules	Detailed breakdown of all 19 system modules with feature specifications
Section 3	Implementation Timeline	Sprint-by-sprint development plan with milestones and deliverables over 13 weeks
Section 4	Training Plan	Training approach, session agenda, and in-app training materials
Section 5	Support & Maintenance	Post-launch support phases, SLA, and backup/recovery strategy
Section 6	Payment Schedule	Milestone-based payment structure with timeline
Section 7	Project Quotation	Consolidated cost summary, inclusions, and acceptance terms

Section 1: Scope of Work

This section defines the complete scope of work for the Permitto PTW Digital Management System, including all deliverables, the permit workflow state machine, items out of scope, and project assumptions.

Project Title	Permitto — Permit To Work (PTW) Digital Management System
Client	PowerChina HDEC
Objective	Design, develop, and deploy a web-based Permit To Work application that digitalizes the client's existing PTW process — replacing paper-based permit management with a multi-level digital approval workflow, SIMOPs coordination, HSSE compliance, real-time tracking, and mobile accessibility.

1.1 In-Scope Deliverables

The following 20 deliverables constitute the complete scope of the project:

#	Deliverable	Description
1	User Management & Authentication	Registration, email/password authentication, JWT-based sessions, password reset. Role-based access control for 5 user roles: Super Admin, PTW-PC (Coordinator), WPI (Issuer), HSSE Officer, WPR (Receiver)
2	Project & Zone Management	Create/manage projects with zone-based divisions. Permits, users, and workflows scoped to specific zones within a project
3	PTW Register	Centralized, auto-maintained digital permit registry — searchable, filterable log of all issued, active, closed, and cancelled permits with full status history
4	Permit Template Builder	Form builder for Coordinators to create custom permit templates for 7 permit types (General PTW, Hot Work, Excavation, Work at Height, Lifting, Night Work, Road Closure) with configurable field types, risk assessment checklists, and compliance sections
5	Permit Request & Submission	Multi-step form workflow for WPR to verify risk assessment controls, fill PTW forms, auto-save during filling, upload supporting documents, and submit to WPI
6	4-Level Approval Workflow	Sequential approval chain: WPR → WPI → PTW-PC → HSSE → APPROVED. Each level includes digital signature, risk assessment verification, and ability to request additional measures (revise & resubmit loop)
7	SIMOPs Management	Simultaneous Operations gate — automatic detection of overlapping permits in the same zone, countersign requirement from other SIMOP's

		WPI, permit withholding mechanism
8	SIMOPs Checklist	Digital SIMOPs checklist (based on client-provided format) for documenting interface management between concurrent operations
9	Daily Re-Approval Cycle	Automated daily re-confirmation for all active permits with deadline enforcement, automatic suspension for missed approvals, and bulk approval support
10	Condition Change Workflow	Mid-work re-assessment process — WPR/WPI can flag changed conditions, triggering re-assessment that leads to either continued work, re-permitting, or cancellation
11	Permit Extension	Formal extension request flow — WPR requests extension before expiry, WPI reviews and grants/denies, extension signed and PTW Register updated
12	Permit Close-Out & Cancellation	Formal close-out by PTW-PC with completion notes and signature. Separate cancellation path for permits terminated due to issues
13	PDF Generation	Automated generation of print-ready permit PDFs matching client-provided physical permit formats (7 types) — with form data, signatures, QR code, and company branding
14	Notification System	Email notifications for all workflow events and in-app notification center with real-time updates and unread counts
15	File Storage	Secure server-based document and signature storage with access control
16	Background Automation	Scheduled jobs for daily reminders, suspension checks, expiry warnings, extension deadline alerts, and end-of-day processing
17	Audit Trail	Complete timestamped audit log for every permit action — submissions, approvals, rejections, additional measures requests, condition changes, extensions, and closures
18	Progressive Web App (PWA)	Installable PWA with offline mode — field workers can view active permits, fill forms, and queue submissions while offline, with automatic sync on connectivity restore
19	Responsive Web Application	Mobile-first design accessible via any modern browser on mobile, tablet, and desktop
20	Deployment & Infrastructure	Docker-containerized deployment with CI/CD pipeline, staging and production environments

1.2 Permit Workflow State Machine

The following state machine defines all permit lifecycle transitions:

```
DRAFT → SUBMITTED → [SIMOPs Gate] → WPI_APPROVED → PC_APPROVED → HSSE_APPROVED → ACTIVE


At each approval level, the approver may:
  • Approve — permit advances to the next level
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  • Request Additional Measures — WPR revises and resubmits
  • Reject — permit returned with feedback


During ACTIVE:
  ACTIVE → CONDITION_CHANGED → RE_ASSESSMENT → RE_PERMIT / CONTINUE / CANCEL
  ACTIVE → EXPIRED → EXTENSION_REQUESTED → GRANTED (back to ACTIVE) / DENIED → CLOSE_OUT
  ACTIVE → PENDING_DAILY → APPROVED (daily) / SUSPENDED


Terminal states:  CLOSED (normal completion)  |  CANCELLED (terminated due to issues)
```

1.3 Out of Scope (Future Phases)

The following items are explicitly out of scope for this engagement and may be addressed in future phases:

- Native mobile applications (iOS/Android)
- SMS notifications
- Multi-language / localization support
- Third-party system integrations (ERP, SAP, etc.)
- Analytics dashboard and reporting module
- Bulk user import (CSV)
- Browser push notifications

1.4 Assumptions

- **1.** Client has provided permit PDF templates (7 types + SIMOPs checklist); additional templates will be provided by the client as needed
- **2.** Client will provide company branding assets (logo, colors) for the application UI
- **3.** Client will assign a dedicated team or representative who will be actively available throughout the project for requirements clarification, feedback, design approvals, UAT participation, and go-live sign-off. Delays in client-side responses may impact the delivery timeline
- **4.** Client will provide a VPS server for hosting the application
- **5.** Client will provide a domain name for deployment
- **6.** Client will provide an emailing service for transactional notifications
- **7.** The system will support 100 concurrent users at launch, with architecture designed to scale to 500+
- **8.** All permit content, forms, and workflow definitions will be provided/validated by the client
- **9.** Users will access the system via modern web browsers (Chrome, Safari, Edge, Firefox) — no legacy browser support required

Section 2: System Features & Modules

This section provides a detailed breakdown of all 19 system modules that comprise the Permitto PTW system. Each module is described with its features and specifications.

2.1 User Management & Authentication

Feature	Description
User Registration	Admin-invited registration with email activation
Login / Logout	Email/password authentication with secure session management
Password Reset	Self-service password reset via email link
Session Management	JWT access tokens (short-lived) + refresh tokens (long-lived) with automatic renewal
Role Assignment	5 roles: Super Admin, PTW-PC (Coordinator), WPI (Issuer), HSSE Officer, WPR (Receiver)
User Profiles	View/edit personal details, notification preferences, and one-time digital signature setup
Digital Signature Setup	User draws and saves their signature once during profile setup. Stored securely in the database and reused for all approvals
User Deactivation	Soft-delete — deactivated users cannot login but historical records are preserved

2.2 Role-Based Access Control (RBAC)

Permissions are enforced at both API level (backend middleware) and UI level (feature visibility).

Role	Capabilities
Super Admin	Create projects, manage all users (all roles), system-wide visibility, system configuration
PTW-PC (Coordinator)	Create permit templates, create Issuers & Receivers, 3rd-level approval, close-out permits, manage PTW Register
WPI (Issuer)	1st-level approval, create Receivers, daily re-approval, SIMOPs countersign, grant/deny permit extensions
HSSE Officer	4th-level (final) approval, independent risk assessment verification, request additional safety measures, HSSE compliance oversight
WPR (Receiver)	Request permits, fill multi-step forms, upload documents, daily re-confirmation,

	request permit extensions

2.3 Project & Zone Management

Feature	Description
Project CRUD	Create, view, edit, activate/deactivate projects
Project Code	Unique project code used in permit ID generation (e.g., "AFIF2")
Zone Management	Divide projects into zones/areas — permits are issued per zone
User-Project Assignment	Assign users to specific projects with their role; users only see assigned projects
Zone-Based Scoping	Permits, SIMOPs detection, and workflows scoped to zones within a project

2.4 PTW Register

Feature	Description
Auto-Registration	Every permit automatically logged in the register upon issuance
Search & Filter	Search by permit number, project, zone, type, status, date range, assignee
Status Tracking	Real-time status visibility for all permits (draft through closed)
Register Updates	Automatic updates on every state change — approvals, extensions, suspensions, closures, cancellations
Export	Export register data for reporting and compliance purposes

2.5 Permit Template Builder

Feature	Description
Template Creation	Coordinators build custom permit form templates per permit type
7 Permit Types	General PTW, Hot Work, Excavation, Work at Height, Lifting, Night Work, Road Closure
Field Types	Text, textarea, number, date, time, select, multi-select, checkbox, radio, file upload, signature, checkbox grid, yes/no/NA
Multi-Step Sections	Organize forms into logical steps/sections with progress indication
Risk Assessment Checklists	Built-in compliance checklist sections with configurable items
Field Properties	Label, placeholder, required/optional, validation rules, help text
Template Versioning	Version control — new versions don't break existing permits using older versions
Template Preview	Preview template layout before publishing

Background Color	Configurable background color per template for visual identification
Activate/Deactivate	Control which templates are available for new permit requests

2.6 Permit Request & Submission

Feature	Description
Template Selection	WPR browses available templates filtered by project and permit type
Multi-Step Form	Step-by-step form filling with progress indicator and forward/backward navigation
Auto-Save	Automatic draft saving during form filling — no data loss on accidental navigation or connectivity issues
File Upload	Upload supporting documents (PDF, JPG, PNG, HEIC) up to 25MB per file, max 10 per permit
Risk Assessment Verification	WPR confirms implementation of risk assessment control measures at site before submission
Draft Management	Save, resume, and manage draft permits before submission
Permit ID Generation	Auto-generated structured permit IDs: {PROJECT_CODE}-{TYPE}-{YEAR}-{SEQUENCE}
Submit to WPI	One-click submission triggers the approval workflow and notifies the assigned WPI

2.7 4-Level Approval Workflow

Approval Level	Actor	Actions
Level 1	WPI (Issuer)	Review permit, verify risk assessment, approve / request additional measures / reject
Level 2	PTW-PC (Coordinator)	Review permit, approve / reject
Level 3	HSSE Officer	Independent risk assessment verification, approve / request additional measures / reject

Feature	Description
One-Click Approval	On approval, user clicks "Approve" and their stored profile signature is automatically applied with a fresh timestamp
Request Additional Measures	Approvers can request specific additional safety measures from WPR — not just flat reject. WPR implements and resubmits
Rejection with Feedback	Mandatory reason/comments on rejection — WPR receives feedback and can revise & resubmit

Approval Dashboard	Each role sees their pending approvals with filters by project, zone, type, urgency
Approval Notifications	Real-time notifications at each workflow transition to all relevant parties
Audit Logging	Every approval, rejection, and additional measures request logged with timestamp, user, and comments

2.8 SIMOPs Management

Feature	Description
Automatic Detection	System detects when multiple permits are active or pending in the same zone during overlapping time periods
SIMOPs Gate	Blocking step in approval flow — if SIMOPs detected, permit cannot proceed until interfaces are managed
Countersign Requirement	Other SIMOP's WPI must review and countersign the permit, confirming interface management
Withhold Mechanism	Permits withheld if interface management issues are unresolved — WPI notified to resolve before proceeding
SIMOPs Checklist	Digital checklist (based on client-provided format) documenting how interfaces between concurrent operations are managed
Cross-Permit Visibility	All parties can see which other permits are active in the same zone

2.9 Daily Re-Approval Cycle

Feature	Description
Daily Reminders	Automated reminder notifications sent at configurable time (default 7:00 AM)
One-Click Re-Approval	User confirms permit validity for the day — stored signature auto-applied with timestamp
Deadline Enforcement	Configurable deadline (default 8:00 AM) — permits not re-approved by deadline are automatically suspended
Bulk Approval	Approve multiple permits at once for efficiency
Suspension	Automatic suspension with notifications to all parties when any role misses the deadline
Reactivation	Suspended permits can be reactivated once all pending daily approvals are completed

2.10 Condition Change & Re-Assessment

Feature	Description
Flag Condition Change	WPR or WPI can flag that site conditions have changed during active work
Notify WPI	Automatic notification to WPI when conditions are flagged
Re-Assessment	WPI reviews the changed conditions and determines next steps
Re-Permitting	If required, WPI triggers re-permitting — original permit cancelled, new permit initiated
Continue Work	If conditions are manageable, WPI confirms and work continues under existing permit
Cancellation	If conditions are unsafe, WPI cancels the permit immediately

2.11 Permit Extension

Feature	Description
Extension Request	WPR requests extension before permit expiry with justification
WPI Review	WPI reviews extension request and grants or denies
Extension Sign-Off	Granted extensions require WPI approval with stored signature and updated validity dates
Register Update	PTW Register automatically updated with extension details
Denial Path	Denied extensions proceed to permit close-out

2.12 Permit Close-Out & Cancellation

Feature	Description
Close-Out	PTW-PC formally closes permit with completion notes and signature
Cancellation	Separate path for permits terminated due to safety issues, condition changes, or re-permitting
Register Update	Both close-out and cancellation automatically update the PTW Register
Final PDF	Updated PDF generated on-demand reflecting close-out details
File Archival	All permit data and documents preserved in the system for compliance

2.13 PDF Generation

Feature	Description

On-Demand Generation	PDFs are generated on-the-fly when user clicks "View PDF" or "Download PDF" — no PDF files permanently stored on the server
7 Permit Formats	PDF layouts matching client-provided physical permit formats for each permit type
Content	Form data, all approval signatures with timestamps, risk assessment details, QR code linking to digital permit
Branding	Company header, logo, and branding on all generated PDFs
Two-Page Layout	Front page (permit details, signatures) and back page (additional precautions, daily re-validation table)
Download & Print	One-click PDF download or in-browser print from the permit detail view
Offline Availability	Generated PDFs are cached by the PWA for offline viewing

2.14 Notification System

Feature	Description
Email Notifications	Transactional emails for: submission, approval, rejection, additional measures, daily reminders, suspension, expiry warning, extension decisions, close-out
In-App Notifications	Real-time notification center with bell icon and unread count
Click-to-Navigate	Click any notification to go directly to the relevant permit
Mark Read/Unread	Mark individual or all notifications as read
Notification Preferences	Users can configure email notification preferences

2.15 File Storage

All file storage is self-contained on the client-provided VPS — no external cloud storage services required.

Feature	Description
Server-Side Storage	Supporting documents stored securely on the VPS filesystem in access-controlled directories
Secure Upload	Files uploaded through the API with authentication and authorization checks
Supported Formats	PDF, JPG, PNG, HEIC
Size Limits	25MB per file, 10 files per permit
Access Control	File access respects permit visibility permissions — only authorized users can download
Signature Storage	User signatures stored as encrypted data in the database (PostgreSQL) — not as separate files
PDF Storage	Not applicable — PDFs are generated on-demand and cached client-side by the

	PWA

Storage Architecture

Data Type	Storage Location	Notes
User signatures	PostgreSQL database	One-time profile signature, stored securely, reused on all approvals
Permit data	PostgreSQL database	All form data, status, audit logs, timestamps
Supporting documents	VPS filesystem	Uploaded files stored in secure server directories
PDFs	Not stored	Generated on-demand, cached by PWA for offline
Offline form drafts	Browser (IndexedDB)	Encrypted, synced to server on reconnect

No External Storage Required

No external storage services (Cloudflare R2, AWS S3, etc.) are required. The entire system runs self-contained on the client-provided VPS.

2.16 Background Automation

Job	Schedule	Description
Daily Reminder	7:00 AM	Send re-approval reminders to all parties with active permits
Suspension Check	8:05 AM	Automatically suspend permits with missed daily approvals
Expiry Warning	24h before	Notify all parties of upcoming permit expiry
End-of-Day	5:00 PM	Process day-end status transitions
Extension Alerts	As needed	Notify when extension deadlines approach
Email Sending	On trigger	Process queued transactional emails

2.17 Audit Trail

Feature	Description
Comprehensive Logging	Every permit action logged: submissions, approvals, rejections, additional measures, condition changes, extensions, suspensions, closures, cancellations
Timestamped	Precise timestamp on every log entry
User Attribution	Every action attributed to the user who performed it
Status History	Previous status to new status recorded on every transition
Comments	Approver/rejector comments captured in the audit log

Offline Action Logging	All actions performed offline are logged with timestamps and synced to the audit trail on reconnect
Timeline View	Visual timeline of all permit events on the permit detail page

2.18 Progressive Web App (PWA) & Offline Mode

Offline Capabilities

Feature	Description
Installable	Users can install the app on mobile/desktop home screen — works like a native app
Offline Permit Viewing	View active permits and permit details without internet connectivity
Offline PDF Viewing	PDFs for active permits are pre-cached when online and available for viewing/printing offline
Offline Form Filling	Fill permit forms offline — data stored locally in encrypted browser storage
Queue Submissions	Submissions and approvals queued when offline, with visual indicator of pending actions
Auto-Sync	Automatic synchronization when connectivity is restored — queued actions are submitted to the server
Sync Status Indicator	Visual indicator showing online/offline status and number of pending sync actions
Smart Caching	Only the user's own active permits and their PDFs are cached — not all system data
Cache Refresh	When permit data changes, cache is refreshed on next online access

Offline Security Measures

All offline cached data is protected with industry-standard security measures, following the same model used by enterprise applications (SAP Fiori, Microsoft Teams, SafetyCulture).

#	Security Measure	Description
1	HTTPS Only	All data in transit is encrypted. Service workers only function over HTTPS
2	Encryption at Rest	All cached data in browser storage (IndexedDB) encrypted using Web Crypto API with session-derived key
3	Session-Bound Cache	Cached data tied to authenticated session. When session expires, encryption key is destroyed
4	Wipe on Logout	All local data (IndexedDB, Cache API, service worker cache) completely wiped on logout
5	Session Timeout	Auto-lock after configurable idle period (15-30 min). Re-authentication required

		even offline
6	Data Minimization	Only user's own active permits cached. Closed/expired permit data automatically purged
7	Auto-Expiry	Cached data expires automatically after configurable period (24-48 hours)
8	Offline Action Audit	All offline actions logged with timestamps. Server validates on sync

Optional Enhanced Security (Available on Request)

#	Security Measure	Description
9	Remote Wipe	Admin can flag a user's session for device wipe. On next sync, all local data is cleared
10	Biometric Lock	Require device fingerprint or face ID before accessing offline cached data

2.19 Deployment & Infrastructure

All infrastructure runs self-contained on the client-provided VPS — no external cloud services required beyond the email service.

Feature	Description
Docker Containerization	Application packaged in Docker containers for consistent, reproducible deployment
CI/CD Pipeline	Automated build, test, and deployment pipeline
Staging Environment	Pre-production environment for client UAT and testing
Production Environment	Live production deployment on client-provided VPS
Database	PostgreSQL for primary data storage
Cache & Queue	Redis for session management, background job queues, and caching
File Storage	VPS filesystem for supporting document uploads
Backup Strategy	Automated database backups and file system backups on the VPS

Infrastructure Cost Summary

Service	Provider	Cost to Client
VPS Server	Client-provided	Included
Domain	Client-provided	Included
Email Service	Client-provided	Included
Database (PostgreSQL)	Runs on VPS	No additional cost

Cache (Redis)	Runs on VPS	No additional cost
File Storage	VPS filesystem	No additional cost
Cloud Storage (R2/S3)	Not required	None
PDF Storage	Not required	None

Section 3: Implementation Timeline

This section outlines the sprint-by-sprint development plan for the 13-week project delivery, following Agile Scrum methodology with regular sprint reviews and live staging demos.

3.1 Methodology

Framework	Agile Scrum
Sprint 1	3 weeks (working prototype delivery)
Subsequent Sprints	2 weeks each
Sprint Review/Demo	End of each sprint — live demo on staging URL
Staging	Hosted on our infrastructure throughout development. Client receives a live staging URL from Sprint 1
Production	Deployed to client's VPS only after UAT sign-off

3.2 Sprint 1 — Working Prototype (Weeks 1-3)

Deliverable	Details
Project Infrastructure	Monorepo setup, Docker containerization, CI/CD pipeline, staging environment on our VPS
Database Design & Setup	Full relational schema — users, projects, zones, templates, permits, daily approvals, audit logs, notifications
Authentication & Sessions	Email/password login, JWT sessions with refresh tokens, password reset, session management
RBAC (5 Roles)	Super Admin, PTW-PC, WPI, HSSE Officer, WPR — permissions enforced at API and UI level
User Management	Create, view, edit, deactivate/reactivate users with role assignment
Project & Zone Management	Create projects with unique codes, divide into zones, assign users to projects
One-Time Profile Signature	Users draw and save signature once during profile setup, stored securely in database
Permit Template Builder	Form builder with 14 field types, multi-step sections, risk assessment checklists, versioning, preview
Permit Request & Submission	Multi-step form filling with auto-save, file upload, draft management
4-Level Approval Workflow	Full sequential approval: WPR → WPI → PTW-PC → HSSE with stored signatures

Daily Re-Approval Cycle	Daily re-confirmation with deadline enforcement, automatic suspension, bulk approval
Permit Close-Out	Formal close-out by PTW-PC, separate cancellation path
PTW Register	Centralized permit registry — searchable, filterable, with status tracking
Background Automation	Scheduled jobs — daily reminders, suspension checks, expiry warnings
Email & In-App Notifications	Transactional emails + notification center with bell icon
On-Demand PDF Generation	Print-ready permit PDFs with form data, signatures, QR code, branding
Audit Trail	Complete timestamped log of all permit actions
Responsive Frontend	Mobile-first web application with role-based dashboards and full workflow UI
Staging Deployment	Live staging URL delivered to client

Sprint 1 Milestone: WORKING PROTOTYPE

Client receives a live staging URL with the full core PTW workflow operational — including 4-level approval, daily re-approval, template builder, PTW register, notifications, and PDF generation.

3.3 Sprint 2 — SIMOPs, Condition Change & Extensions (Weeks 4-5)

Deliverable	Details
SIMOPs Full Implementation	Countersign flow (other SIMOP's WPI must approve), withhold mechanism, blocking gate in approval workflow
SIMOPs Checklist	Digital checklist based on client-provided format for documenting interface management
SIMOPs UI	Conflict alerts, countersign screens, cross-permit visibility per zone
Condition Change Workflow	Flag changed conditions, notify WPI, re-assessment with options: continue, re-permit, or cancel
Permit Extension	Extension request by WPR, WPI review and grant/deny, sign extension, PTW Register auto-update
Condition Change & Extension UI	Request forms, approval screens, status indicators

3.4 Sprint 3 — PDF Templates & Notifications (Weeks 6-7)

Deliverable	Details
7 Permit PDF Formats	PDF layouts matching all client-provided physical permit formats (General PTW, Hot Work, Excavation, Work at Height, Lifting, Night Work, Road Closure)
Updated Email Notifications	New notifications for HSSE approval, SIMOPs countersign, additional

	measures, condition change, extensions
Additional Measures Flow	Specific measures communicated to WPR with actionable notifications
PDF Validation	Side-by-side comparison with physical permit formats for 1:1 parity

3.5 Sprint 4 — PWA & Offline Mode (Weeks 8-9)

Deliverable	Details
PWA Setup	Service worker, app manifest — installable on mobile/desktop home screen
Offline Permit Viewing	Active permits cached and viewable without connectivity
Offline PDF Caching	PDFs pre-cached for offline viewing/printing
Offline Form Filling	Fill permit forms offline, data stored in encrypted local storage
Auto-Sync	Automatic sync of queued actions on connectivity restore
Encryption at Rest	All cached data encrypted using Web Crypto API
Session-Bound Cache	Encryption key destroyed on logout/session expiry
Session Timeout	Auto-lock after idle period, re-authentication required even offline
Data Minimization	Only user's own active permits cached, auto-purge of expired data
Offline Action Audit	All offline actions logged with timestamps, validated on sync

3.6 Sprint 5 — Mobile Optimization & Testing (Weeks 10-11)

Deliverable	Details
Mobile Responsiveness	All pages audited and optimized for mobile and tablet use
Touch Target Optimization	Buttons and interactions sized for field use on mobile
Cross-Browser Testing	Chrome, Safari, Edge, Firefox verification
End-to-End Workflow Testing	All permit types, all approval paths, all edge cases
SIMOPs Edge Cases	Multiple overlapping permits, zone conflict scenarios
Offline Sync Testing	Conflict resolution, queue integrity verification
Performance Optimization	Page load times, API response times
Bug Fixes	All issues from previous sprint reviews

3.7 Sprint 6 — UAT & Production Deployment (Weeks 12-13)

Deliverable	Details
Client UAT	Client team tests all workflows on staging environment
UAT Feedback Fixes	Issues reported during UAT resolved
Final Adjustments	UI tweaks, wording changes, workflow adjustments per client feedback
Client VPS Setup	Docker, PostgreSQL, Redis, domain configuration on client's VPS
Production Deployment	Application deployed to client's VPS
Initial Data Setup	Users, projects, zones, templates configured for production
Production Smoke Test	Full workflow verification on production environment
Handover	Admin credentials, documentation, system access

Sprint 6 Milestone: PROJECT DELIVERED

Production live on client's VPS. UAT signed off. Source code, documentation, and admin access handed over.

3.8 Timeline Summary & Key Milestones

Sprint Overview

Sprint	Duration	Weeks	Deliverable
Sprint 1: Working Prototype	3 weeks	1-3	Live staging URL — full core PTW workflow
Sprint 2: SIMOPs + Extensions	2 weeks	4-5	SIMOPs, condition change, permit extension
Sprint 3: PDF Templates + Notifications	2 weeks	6-7	7 PDF formats, complete notification system
Sprint 4: PWA & Offline Mode	2 weeks	8-9	Installable PWA with secure offline mode
Sprint 5: Mobile + Testing	2 weeks	10-11	Mobile optimization, full testing
Sprint 6: UAT + Production	2 weeks	12-13	Client UAT, production deployment
Total	**13 weeks**		

Key Milestones

Milestone	When	Deliverable

Working Prototype	End of Week 3	Live staging URL — core PTW workflow demo
SIMOPs Complete	End of Week 5	Full SIMOPs with countersign and checklist
All Features Complete	End of Week 9	PWA, offline, all workflows operational
UAT Start	Week 12	Client begins formal acceptance testing
Go-Live	End of Week 13	Production on client's VPS

Hosting Approach

Phase	Environment	Hosted On
Sprints 1-5	Staging	Our VPS — client receives staging URL for sprint reviews
Sprint 6	Production	Client's VPS — final deployment after UAT sign-off

Section 4: Training Plan

This section outlines the training approach to ensure PowerChina HDEC's team is fully equipped to use and manage the Permitto system independently after go-live.

4.1 Training Approach

Method	Single comprehensive train-the-trainer session with client's team
Format	Live session (online or on-site) covering all roles, workflows, and system administration
Timing	Conducted during Sprint 6 (UAT phase), before go-live
Environment	Conducted on the staging environment with test data
Self-Service Materials	Dedicated in-app training & help page accessible by all registered users

4.2 Training Session

Audience	Client's designated team (train-the-trainer)
Duration	3-4 hours
Format	Live walkthrough + Q&A

Session Agenda

#	Topic	Coverage
1	System Overview & Navigation	Login, dashboard, navigation, notification center, PWA installation, offline mode
2	User & Project Setup	Creating users across all 5 roles, project creation, zone management, user-project assignment
3	Profile Signature Setup	One-time signature drawing and saving for all approval roles
4	Template Builder	Creating permit templates, field types, multi-step sections, risk assessment checklists, versioning, preview
5	Template Modification	Editing existing templates, creating new versions, duplicating templates
6	Permit Request Flow (WPR)	Selecting template, filling multi-step form, auto-save, file upload, draft management, submitting

7	4-Level Approval Flow	WPI approval, PTW-PC approval, HSSE approval — one-click approve, requesting additional measures, rejection
8	SIMOPs Management	Zone overlap alerts, countersign process, withhold mechanism, SIMOPs checklist
9	Daily Re-Approval	Daily confirmation, bulk approval, suspension and reactivation
10	Condition Change & Extensions	Flagging condition changes, re-assessment, extension request and approval
11	Close-Out & Cancellation	Permit close-out by PTW-PC, cancellation, PTW Register updates
12	PTW Register	Searching, filtering, exporting permit records
13	PDF Generation	Viewing and downloading permit PDFs, offline PDF access
14	Q&A	Open questions from the client's team

After this session, the client's trained team will be equipped to onboard all end users (WPIs, HSSE Officers, WPRs) independently.

4.3 In-App Training & Help Page

A dedicated Training & Help page will be built into the web application, accessible by all registered users from the main navigation.

Feature	Description
Role-Based Guides	Each user sees training content relevant to their role
Step-by-Step Walkthroughs	Visual guides with screenshots for every major workflow
Video Walkthroughs	Recorded screen recordings embedded in the help page
Quick Reference	Common actions per role — one-page summary format
FAQ Section	Frequently asked questions and troubleshooting steps
Searchable	Users can search for specific topics or actions
Always Accessible	Available from any page via a help icon in the navigation

4.4 Training Materials Summary

Material	Delivery Method
Live train-the-trainer session	Online or on-site (1 session, 3-4 hours)
Role-based user guides	In-app training page
Step-by-step walkthroughs with screenshots	In-app training page
Video recordings of training session	In-app training page
Quick reference per role	In-app training page

FAQ & troubleshooting	In-app training page

All training materials are digital, accessible within the application, and available to every user at any time — no external documents to manage or distribute.

Section 5: Support & Maintenance

This section defines the post-launch support structure, including the intensive hypercare period, ongoing maintenance, service level agreements, and backup/recovery strategy.

5.1 Support Phases

Phase	Period	Description
Hypercare	First 4 weeks after go-live	Intensive post-launch support — priority response to any issues, daily monitoring, immediate bug fixes
Active Support	Ongoing (after hypercare)	Continued maintenance, bug fixes, security updates, server monitoring, and technical assistance

5.2 Hypercare Period (Weeks 1-4 After Go-Live)

Dedicated intensive support immediately after production deployment to ensure smooth adoption.

Duration	4 weeks from go-live date
Availability	Sunday–Thursday, business hours (or as agreed)
Response Time	Critical issues: within 2 hours. Other issues: within 4 hours
Monitoring	Daily system health checks — server, database, background jobs, email delivery
Scope	Bug fixes, performance issues, workflow adjustments, user support, configuration changes
Communication	Direct channel (WhatsApp/Teams/email) with the development team

5.3 Active Support & Maintenance (Ongoing)

Continued support after the hypercare period to keep the system running smoothly, secure, and up to date.

What's Included

Category	Coverage
Bug Fixes	Identification and resolution of software defects — no additional charge for bugs in delivered features
Security Updates	Regular security patches for dependencies, frameworks, and server software

Server Monitoring	Monitoring of application health, database performance, background job execution, and disk usage
Database Maintenance	Routine database optimization, backup verification, and performance tuning
Uptime Monitoring	System availability monitoring with alerts for downtime
Backup Management	Automated daily database backups with verification
SSL Certificate Management	SSL/TLS certificate renewal and configuration
Email Service Monitoring	Monitoring email delivery success rates and troubleshooting delivery issues
Background Job Monitoring	Ensuring scheduled jobs execute reliably
Minor Adjustments	Small configuration changes, wording updates, notification timing adjustments

Service Level Agreement (SLA)

Priority	Description	Response Time	Resolution Target
Critical	System down, no users can access, data loss risk	Within 2 hours	Within 8 hours
High	Major feature broken (approvals, PDFs, notifications)	Within 4 hours	Within 24 hours
Medium	Feature partially impaired, workaround available	Within 8 hours	Within 3 business days
Low	Minor UI issue, cosmetic defect, non-urgent request	Within 24 hours	Within 5 business days

Support Availability

Support Hours	Sunday–Thursday, 9:00 AM – 6:00 PM (client's timezone)
Emergency Support	Critical issues (system down) — available outside business hours via emergency contact
Communication Channels	Dedicated support channel (WhatsApp/Teams/email — as agreed)
Issue Reporting	Client reports issues via agreed channel with description and screenshots

5.4 System Updates & Improvements

Type	Included in Maintenance	Description
Bug Fixes	Yes	Defects in delivered features
Security Patches	Yes	Dependency and server updates

Minor Adjustments	Yes	Configuration, wording, timing changes
Performance Optimization	Yes	Query optimization, caching improvements
New Features / Enhancements	No — quoted separately	New modules, workflow changes, additional permit types
Major Scope Changes	No — quoted separately	Significant functionality additions beyond original scope

New features and enhancements beyond the delivered scope can be requested and will be estimated and quoted separately.

5.5 Backup & Disaster Recovery

Database Backups	Automated daily backups with 30-day retention
Backup Storage	Stored separately from production server for redundancy
Recovery Time Objective (RTO)	System restored within 4 hours of a critical failure
Recovery Point Objective (RPO)	Maximum 24 hours of data loss (from last daily backup)
Disaster Recovery Test	Backup restoration tested quarterly to verify integrity

Section 6: Payment Schedule

This section outlines the milestone-based payment structure for the project development and the monthly retainer for ongoing support and maintenance.

6.1 Payment Structure

Component	Type	Detail
Project Development	Milestone-based	Payments tied to sprint deliverables and client sign-off
Support & Maintenance	Monthly retainer	USD 1,000/month, begins after go-live (minimum 6 months)

6.2 Development Payment Milestones

#	Milestone	Trigger	Amount (USD)
1	Project Kick-Off	Contract signing — before development begins	$4,000
2	Working Prototype Delivered	Sprint 1 complete — live staging URL with core PTW workflow	$1,600
3	SIMOPs & Advanced Workflows	Sprint 2 complete — SIMOPs with countersign, condition change, permit extension	$1,400
4	PDF Templates & Notifications	Sprint 3 complete — 7 permit PDF formats, complete notification system	$1,200
5	PWA & Offline Mode	Sprint 4 complete — installable PWA with secure offline mode	$1,400
6	Testing Complete	Sprint 5 complete — mobile optimization, full testing, bug fixes	$1,200
7	Go-Live & Handover	Sprint 6 complete — production deployed on client's VPS, UAT signed off	$1,200
	Total Project Cost		**$12,000**

6.3 Support & Maintenance Payment

Monthly Fee	USD 1,000/month
Starts	After go-live (end of Sprint 6)

Minimum Term	6 months
Includes	Hypercare (first 4 weeks) + ongoing active support as defined in Section 5
Billing Cycle	Monthly, invoiced at the start of each month
After Minimum Term	Continues month-to-month, cancellable with 30 days written notice

6.4 Total Cost Summary

Item	Amount (USD)
Project Development (one-time)	$12,000
Support & Maintenance (6-month minimum)	$6,000 ($1,000/month x 6)
TOTAL	**$18,000**

6.5 Payment Terms

Currency	USD
Payment Method	Bank transfer / wire (or as agreed)
Invoice	Issued upon milestone completion with deliverable summary
Payment Due	Within 7 business days of invoice date
Milestone Sign-Off	Client confirms deliverable acceptance via sprint review before payment is due
Late Payment	Work may be paused if payment is overdue by more than 14 days

6.6 Payment Timeline Overview

Week	Event	Payment
Week 0	Contract signing	$4,000 (kick-off)
Week 3	Sprint 1 — Prototype delivered	$1,600
Week 5	Sprint 2 — SIMOPs & workflows	$1,400
Week 7	Sprint 3 — PDFs & notifications	$1,200
Week 9	Sprint 4 — PWA & offline	$1,400
Week 11	Sprint 5 — Testing complete	$1,200
Week 13	Sprint 6 — Go-live & handover	$1,200
Week 14+	Support & maintenance begins	$1,000/month

Section 7: Project Quotation

This section provides the consolidated project quotation, including scope overview, cost breakdown, what's included, client responsibilities, and acceptance terms.

7.1 Quotation Summary

Project	Permitto — Permit To Work (PTW) Digital Management System
Prepared For	PowerChina HDEC
Prepared By	[Your Company Name]
Date	[Date]
Validity	This quotation is valid for 30 days from the date above

7.2 Scope Overview

Design, development, and deployment of a web-based Permit To Work application that digitalizes the client's existing PTW process with:

- 5 user roles (Super Admin, PTW-PC, WPI, HSSE Officer, WPR)
- 4-level sequential approval workflow (WPR → WPI → PTW-PC → HSSE)
- 7 permit types matching client-provided physical formats
- SIMOPs (Simultaneous Operations) management with countersign and withhold mechanism
- Zone-based project management
- Daily re-approval cycle with automatic suspension
- Condition change workflow and permit extension flow
- On-demand PDF generation matching physical permit layouts
- Progressive Web App (PWA) with secure offline mode
- In-app training & help page
- Full audit trail and PTW Register
- Deployed on client's VPS with no external service dependencies

7.3 Cost Breakdown

A. Project Development

#	Deliverable	Timeline	Amount (USD)
1	Project Kick-Off	Week 0	$4,000

2	Working Prototype — core PTW workflow, 4-level approval, zones, PTW Register	Week 3	$1,600
3	SIMOPs, Condition Change & Permit Extension	Week 5	$1,400
4	7 Permit PDF Formats & Complete Notification System	Week 7	$1,200
5	PWA & Secure Offline Mode	Week 9	$1,400
6	Mobile Optimization & Full Testing	Week 11	$1,200
7	UAT, Production Deployment & Handover	Week 13	$1,200
	Subtotal — Development	**13 weeks**	**$12,000**

B. Support & Maintenance

Item	Detail	Amount (USD)
Monthly Support & Maintenance	Bug fixes, security updates, server monitoring, database backups, SLA-based support	$1,000/month
Minimum Term	6 months	$6,000
After Minimum Term	Month-to-month, cancellable with 30 days notice	$1,000/month

C. Total

Item	Amount (USD)
Project Development (one-time)	$12,000
Support & Maintenance (6-month minimum)	$6,000
TOTAL	**$18,000**

7.4 What's Included

Category	Details
Full System Development	All 20 deliverables as defined in Sections 1 & 2
Staging Environment	Hosted on our infrastructure throughout development
Production Deployment	Setup and deployment on client's VPS
Training	1 comprehensive train-the-trainer session (3-4 hours)
Training Materials	In-app training & help page with role-based guides, walkthroughs, videos, and FAQ
Hypercare	4 weeks of intensive post-launch support (included in maintenance)

Source Code	Full source code delivered to client at handover
Documentation	System documentation and admin guide

7.5 What Client Provides

Item	Detail
VPS Server	For production deployment
Domain Name	For production URL
Email Service	For transactional notifications
Branding Assets	Company logo and brand colors
Permit Templates	Physical permit formats (provided) and additional formats as needed
SPOC / Team	Dedicated team or representative for feedback, approvals, and UAT

No Additional Infrastructure Costs

Cloud Storage (R2/S3): Not required | PDF Storage: Not required (on-demand generation) | Database: Runs on client's VPS | Redis Cache: Runs on client's VPS — No additional cost.

7.6 Payment Terms

Currency	USD
Kick-Off Payment	$4,000 due upon contract signing
Milestone Payments	Due within 7 business days of milestone sign-off
Support Payments	$1,000/month, invoiced at the start of each month
Payment Method	Bank transfer / wire (or as agreed)

7.7 Acceptance & Signatures

This quotation is accepted by the authorized representative of the client:

CLIENT

Client Name	
Authorized Signatory	
Designation	

Signature	
Date	

PROVIDER

Provider Name	
Authorized Signatory	
Designation	
Signature	
Date	